

ELEC 221 2026W1 Tutorial 2

Monday 21 September 2026

Suggested Oppenheim problems: 1.16-1.19, 1.22e-g, 2.1-2.7, 2.21

1 Linearity and time invariance

Recall that a linear system $x(t) \rightarrow y(t)$ satisfies additivity and homogeneity:

$$ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t), \quad \text{for arbitrary (possibly complex) } a, b.$$

If we input a weighted sum of signals to a linear system we get a weighted sum of output signals with the same weights.

A system is time invariant if a time-shifted input produces an identically time-shifted output:

$$x(t) \rightarrow y(t) \implies x(t - t_0) \rightarrow y(t - t_0), \quad \text{for every } t_0.$$

Intuitively, if we input the same signal into a time-invariant system tomorrow, we'll get the same output as we'd get today.

A system that is both linear and time invariant is called LTI.

1. (Oppenheim 1.19 variant) For each of the following input-output relationships, determine whether the corresponding system is linear, time invariant, or both.

(a) $y(t) = t^2 x(t - 1)$

(b) $y[n] = x[n + 1] - x[n - 1]$

(c) $y(t) = \sin(x(t))$

2 The unit impulse, the unit step, and the impulse response

The DT unit impulse and DT unit step are defined by

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}, \quad u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}.$$

They are related by the following expressions:

$$\delta[n] = u[n] - u[n - 1], \quad u[n] = \sum_{m=-\infty}^n \delta[m].$$

The impulse can “sift out” or “sample” a signal at specific points:

$$x[n] \delta[n - k] = x[k] \delta[n - k].$$

This enables us to write *any* signal as a superposition of weighted, shifted impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k].$$

This might look redundant at first. Recall in class we made an analogy with writing a vector as a weighted sum of basis vectors.

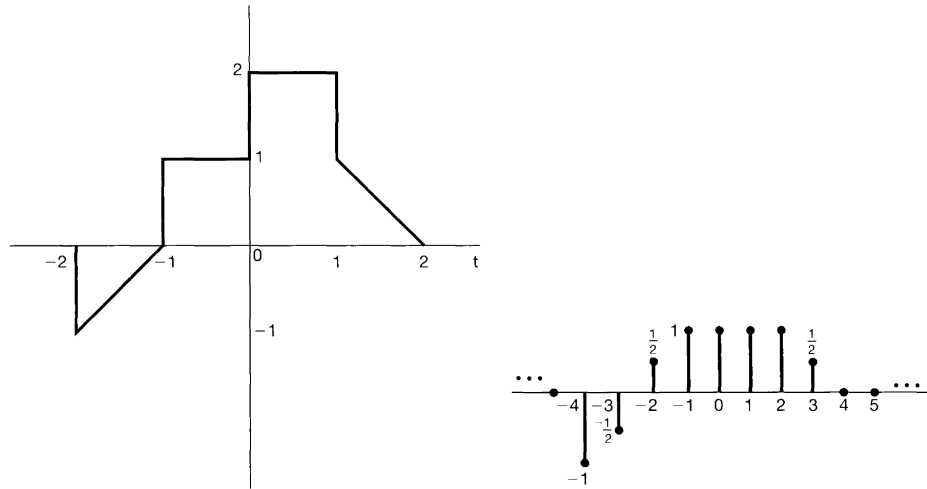


Figure 1: Image credit: Figures P1.21 and P1.22 from Oppenheim (p. 60).

1. (Oppenheim 1.22) A DT signal $x[n]$ is shown in Figure 1. Sketch and label the following signals:

- (a) $x[n] u[3 - n]$
- (b) $x[n - 2] \delta[n - 2]$
- (c) $\frac{1}{2}x[n] + \frac{1}{2}(-1)^n x[n]$

3 The convolution sum

With this decomposition we can reason about LTI systems. If an LTI system sends $\delta[n] \rightarrow h[n]$, where $h[n]$ is the impulse response, its response to an arbitrary input is

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k] \quad \rightarrow \quad y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k].$$

The operation $*$ is called convolution, and the expression on the right is called the convolution sum.

1. (Based on Oppenheim 1.16) Consider a DT system with input $x[n]$ and output

$$y[n] = x[n] x[n - 2].$$

- (a) (Review) Is the system memoryless? Causal? Stable? Invertible?
- (b) Is the system LTI?

- (c) Determine the output of the system when the input is $A\delta[n]$, where A can be real or complex.
2. Consider a system with impulse response $h[n] = \delta[n] + \frac{1}{2}\delta[n-2]$.
- (a) Sketch the impulse response, and discuss with a classmate: what kinds of physical phenomena or systems might have an impulse responses with similar properties?
- (b) Determine the output of the system when the input is $x[n] = \left(\frac{1}{2}\right)^n u[n]$.
3. Consider a system with impulse response $h[n] = \left(\frac{1}{2}\right)^n u[n]$. Determine the system output if the input signal is $x[n] = \delta[n] + \frac{1}{2}\delta[n-2]$. Explicitly evaluate the convolution sum, but leverage the properties of convolution to check your work. Then, use this to re-express the convolution sum with a time-shifted $x[n]$ rather than $h[n]$.
4. (Oppenheim 2.3) Consider an input $x[n]$ and a system with impulse response $h[n]$ given by

$$x[n] = \left(\frac{1}{2}\right)^{n-2} u[n-2], \quad h[n] = u[n+2].$$

Determine and plot the output $y[n] = x[n] * h[n]$.

5. (Oppenheim 2.4) Compute and plot $y[n] = x[n] * h[n]$, where

$$x[n] = \begin{cases} 1, & 3 \leq n \leq 8 \\ 0, & \text{otherwise} \end{cases}, \quad h[n] = \begin{cases} 1, & 4 \leq n \leq 15 \\ 0, & \text{otherwise} \end{cases}.$$

6. (Oppenheim 2.7) A linear system S has the relationship

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] g[n-2k]$$

between its input $x[n]$ and its output $y[n]$, where $g[n] = u[n] - u[n-4]$.

- (a) Determine $y[n]$ when $x[n] = \delta[n-1]$.
- (b) Determine $y[n]$ when $x[n] = u[n]$.
7. (Oppenheim 2.21) Compute the convolution $y[n] = x[n] * h[n]$ of the following pairs of signals:
- (a) $x[n] = \alpha^n u[n]$, $h[n] = \beta^n u[n]$, with $\alpha \neq \beta$
- (b) $x[n] = h[n] = \alpha^n u[n]$
- (c) $x[n] = \left(-\frac{1}{3}\right)^n u[n-4]$, $h[n] = 4^n u[2-n]$